A08 AIR QUALITY MONITORING AND AWARENESS | O (MAX : 2 PT)

Intent : Monitor indoor air quality issues, as well as inform and educate individuals on the quality of the indoor environment.

Summary : This WELL feature requires the ongoing measurement of contaminant data to educate and empower occupants about their environmental quality.

Issue : Types and concentrations of indoor pollutants continuously fluctuate in any indoor or outdoor environment. For example, cooking in the home can lead to a rapid spike in indoor air pollution.[1] Urban rush hours and waste-burning cause spikes in air pollution outdoors, which can directly impact indoor air quality. Some indoor air pollutants are recognized by their immediate impacts on our body, such as throat irritation or watery eyes.[2,3] However others that go undetected, are not necessarily benign. According to the U.S. Environmental Protection Agency, some health impacts like respiratory diseases, heart disease and cancer can appear years after exposure.[4]

Solutions : Due to air quality fluctuations, it is important to install air quality sensors and detectors in every building. Because air quality can fluctuate throughout the day in every building, real-time monitoring is necessary to promptly fix any deviations in indoor quality metrics and minimize occupant exposure to pollutants. In addition to having robust and calibrated sensors, positioning them correctly plays a crucial role in the accurate assessment of air quality.Furthermore, educating occupants about the risks associated with elevated air pollutant exposures, along with actions they can take to reduce these risks, can encourage personal agency to seek out opportunities to further curb indoor pollution levels.

PART 1 INSTALL INDOOR AIR MONITORS (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Sensor requirements

The project deploys monitors with sensors that measure at least three of the following parameters in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook:

a. $PM_{2.5}$ or PM_{10}.
b. Carbon dioxide.
c. Carbon monoxide.
d. Ozone.
e. Nitrogen dioxide.
f. Total VOCs.
g. Formaldehyde.

2: Reporting & maintenance

The following requirement is met:

a. Proof of calibration or replacement is submitted annually in accordance with the requirements of the WELL Performance Verification Guidebook.

For Dwelling Units & Guest Rooms:

1: Sensor requirements

The project deploys monitors with sensors that measure at least three of the following parameters in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook:

a. $PM_{2.5}$ or PM_{10}.
b. Carbon dioxide.
c. Carbon monoxide.
d. Ozone.
e. Nitrogen dioxide.
f. Total VOCs.
g. Formaldehyde.

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 PROMOTE AIR QUALITY AWARENESS (MAX : 1 PT)

For All Spaces:

Note : Projects may only receive points for this part, if Part 1 is also achieved

The air quality data measured in Part 1 of this feature is made available to occupants as follows:

a. Data are presented through one of the following:
   1. Display screens prominently positioned at a height of 1.1–1.7 m with at least one display per 500 m^2 of regularly occupied space.
   2. Hosted on a website or phone application accessible to occupants. Signs are present indicating where the data

may be accessed at a density of at least one sign per 500 m^2 of regularly occupied space.

b. Data presented include one of the following:
   1. Concentrations of the parameters measured.
   2. Qualitative results of air quality (e.g., colored-coded levels).

c. Data are updated at least once every 15 minutes.

REFERENCES

1. Wallace LA, Emmerich SJ, Howard-Reed C. Source Strengths of Ultrafine and Fine Particles Due to Cooking with a Gas Stove. Environ Sci Technol. 2004;38(8):2304-2311. doi:10.1021/es0306260
2. Joshi S. The sick building syndrome. Indian J Occup Environ Med. 2008;12(2):61. doi:10.4103/0019-5278.43262
3. Selgrade MK, Plopper CG, Gilmour MI, Conolly RB, Foos BSP. Assessing the health effects and risks associated with children's inhalation exposures - Asthma and allergy. J Toxicol Environ Heal - Part A Curr Issues. 2008;71(3):196-207. doi:10.1080/15287390701597897
4. U.S. Environmental Protection Agency. Introduction to Indoor Air Quality. https://www.epa.gov/indoor-air-quality-iaq/introduction-indoor-air-quality.